EP Network-Side Optimization

Isolate, spread, and offload expert-parallel RDMA traffic

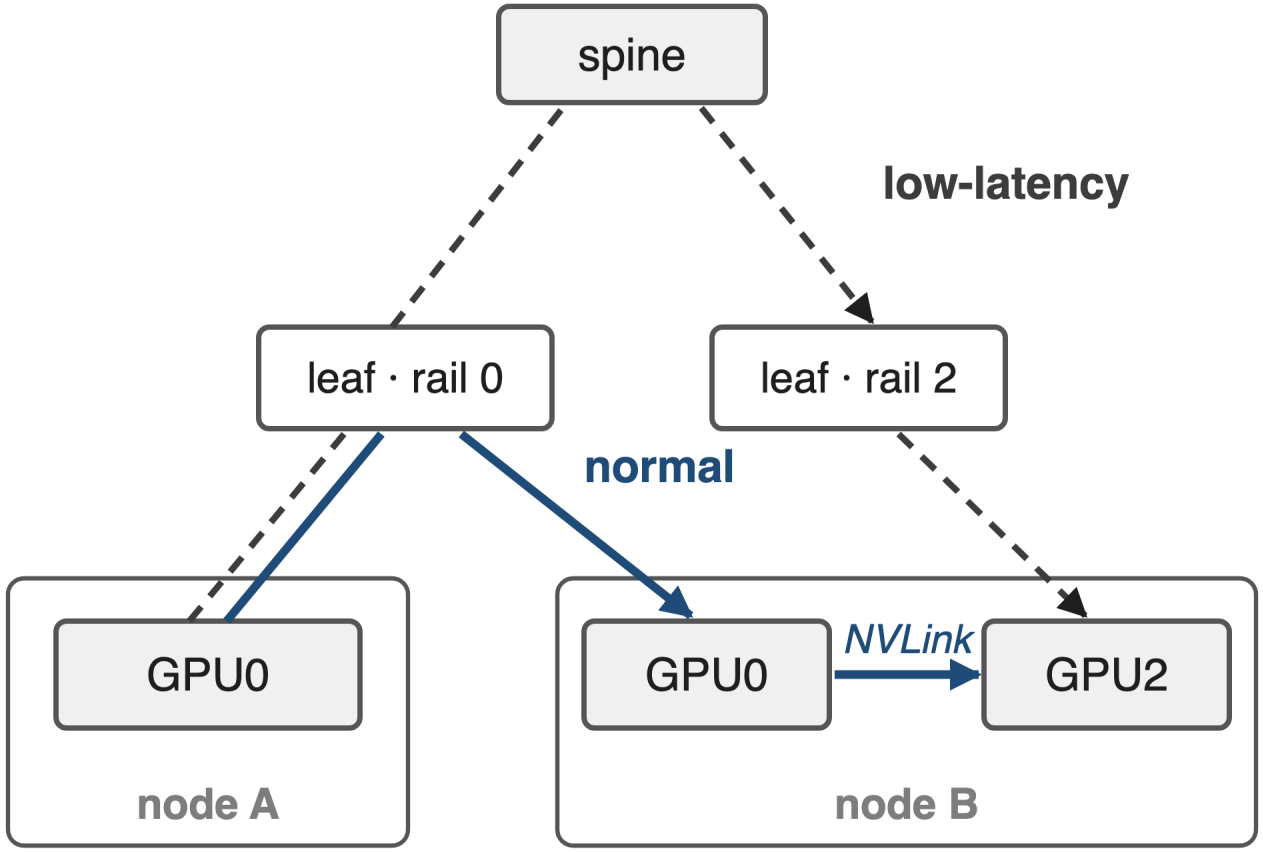
DeepSeek-AI · DeepEP, with AntGroup community branches

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Expert-Parallel Communication · github.com/deepseek-ai/DeepEP

EP crosses the fabric on two paths

- Normal (training, prefill): same rail, then NVLink
- Low-latency (decode): direct RDMA, often via the spine
- H800: IB **50** GB/s vs NVLink **160** GB/s per GPU



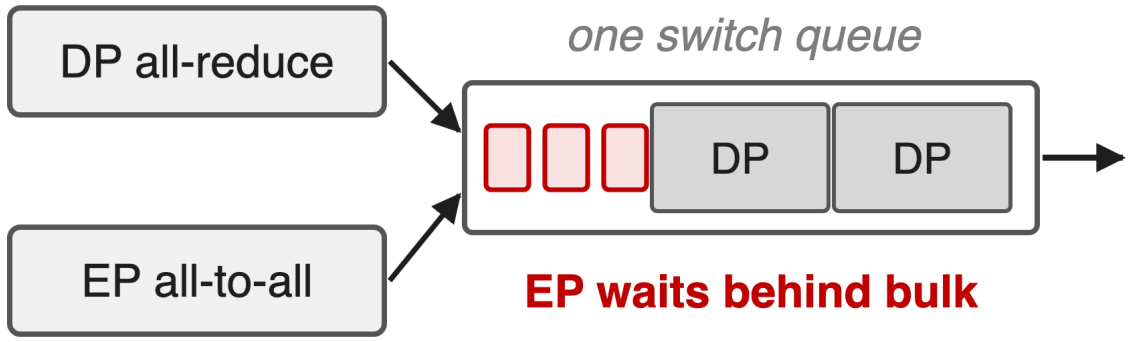
V3 spends 20 of 132 SMs on communication

20 / 132

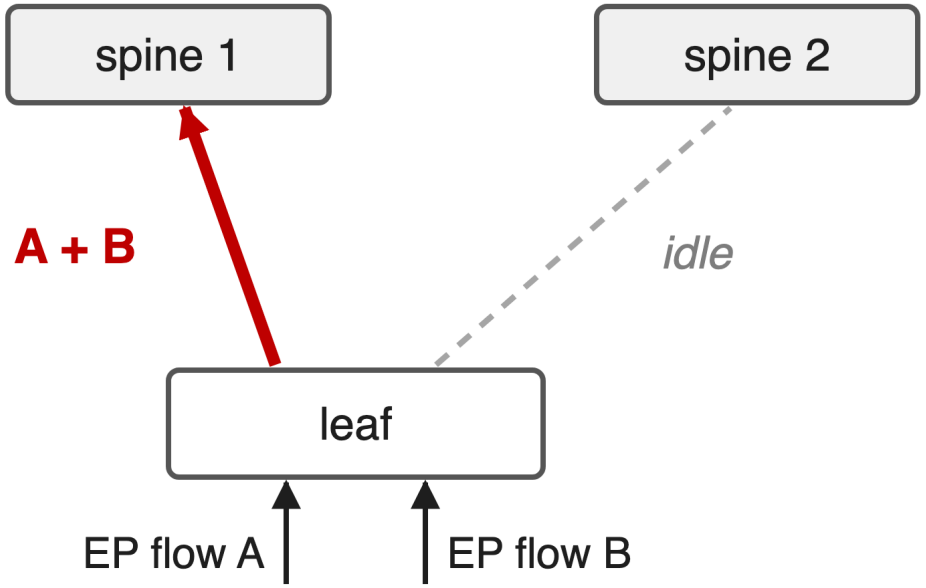
H800 SMs DeepSeek-V3 reserves to drive EP traffic: forwarding, buffer copies, reductions, not math (15%)

DeepSeek-V3 Technical Report §3.5.1 (p.20); ISCA'25 "Insights into DeepSeek-V3" §4.4.1

Shared lanes and hashed paths stall EP packets



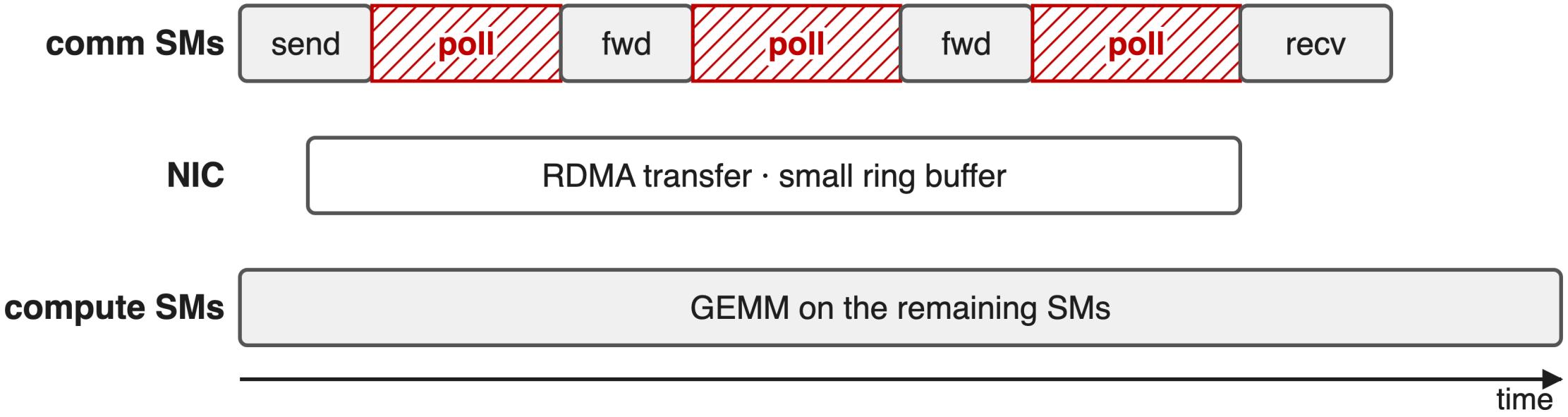
(a) one lane for all traffic



(b) static hash routing

An all-to-all finishes only when its slowest packet lands

Comm SMs spend the transfer polling



A small, reused ring buffer makes SMs wait for free slots

Insight: all-to-all is as slow as its last packet

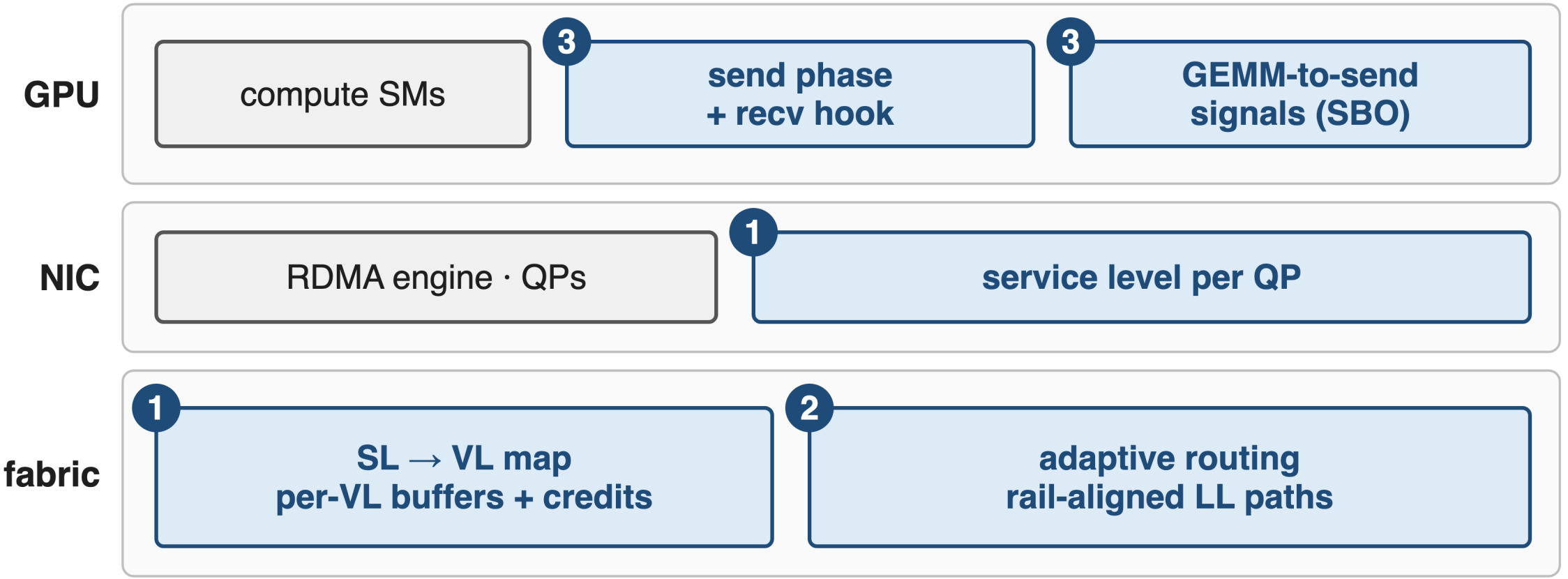
- One blocked or collided packet stalls the layer
- RDMA is one-sided: the NIC needs bytes, not SMs

Insight: all-to-all is as slow as its last packet

- One blocked or collided packet stalls the layer
- RDMA is one-sided: the NIC needs bytes, not SMs

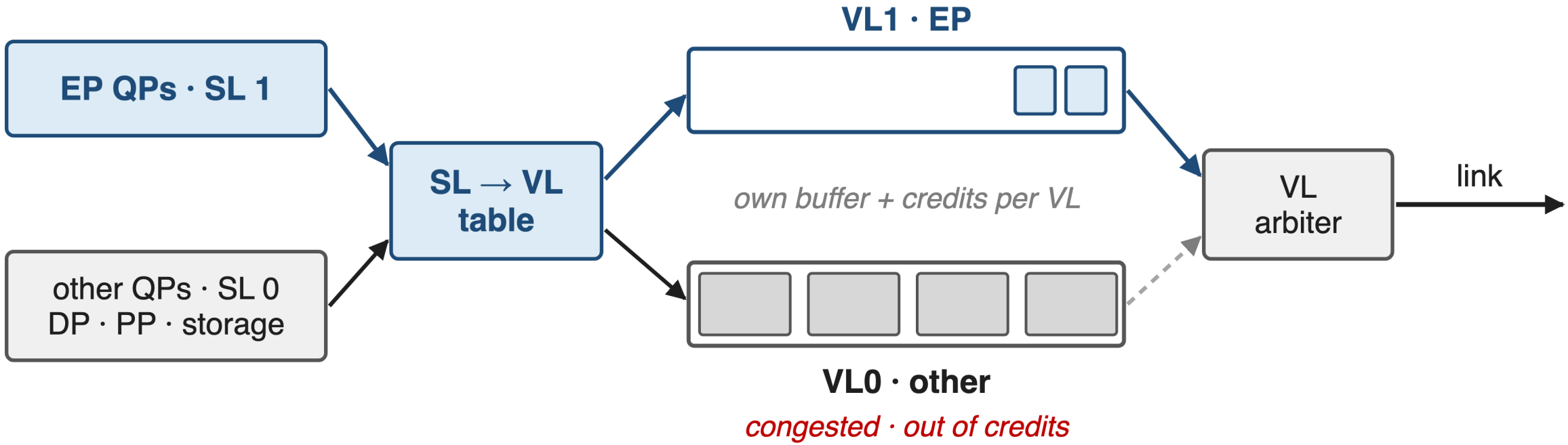
Give EP its own lane and every path; hand bytes to the NIC and step away

Three fixes at three layers



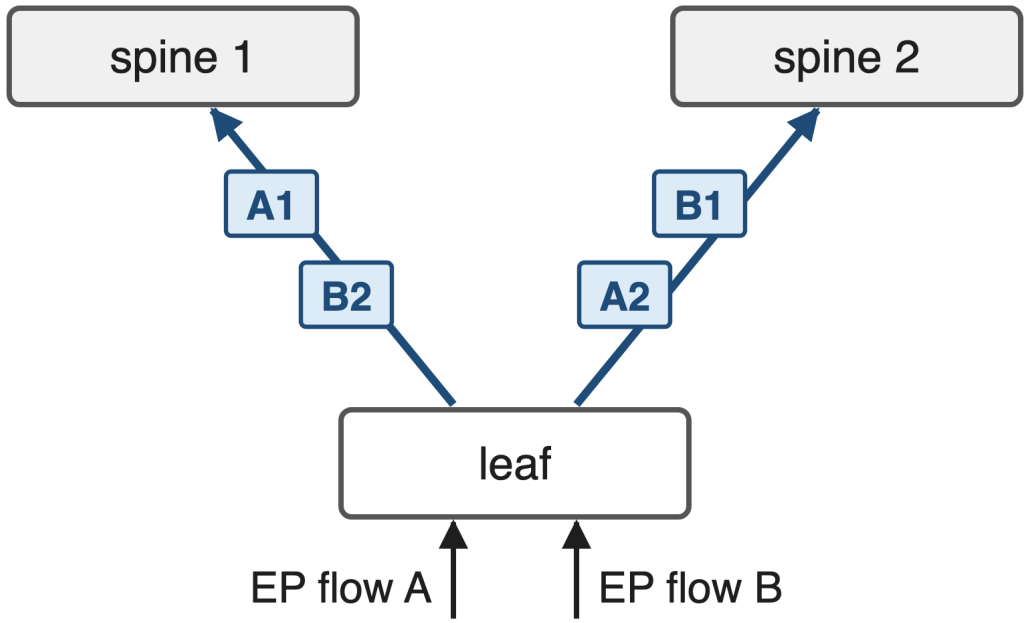
❶ traffic isolation and ❷ adaptive routing: DeepEP README; LL paths and ❸: AntGroup-Opt branch (PR #500, #347, #483)

❶ A service level puts EP on its own VL

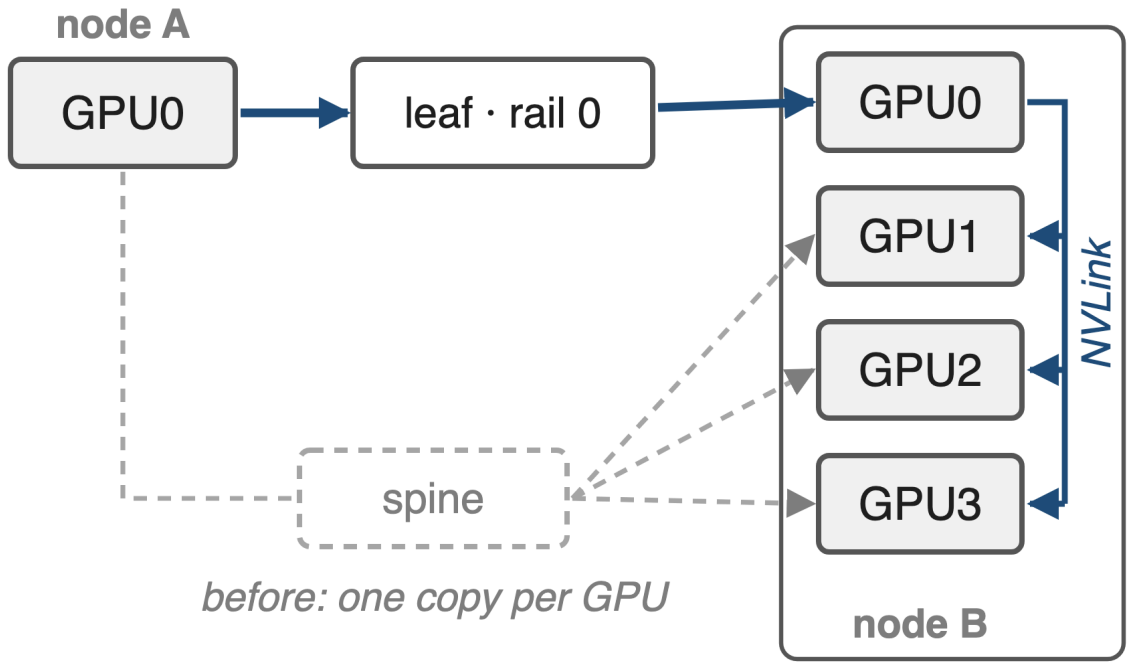


Each VL has its own buffers and credits, so congestion in VL0 never blocks VL1

❷ Spread EP over all uplinks, or skip the spine



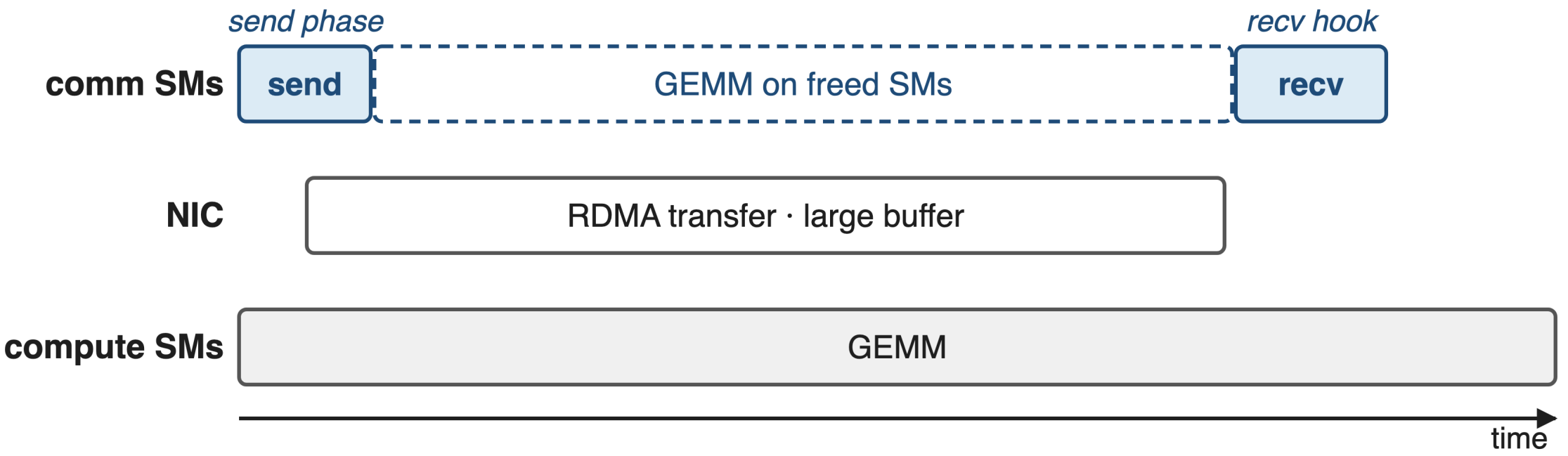
(a) adaptive routing per packet



(b) rail-aligned LL dispatch

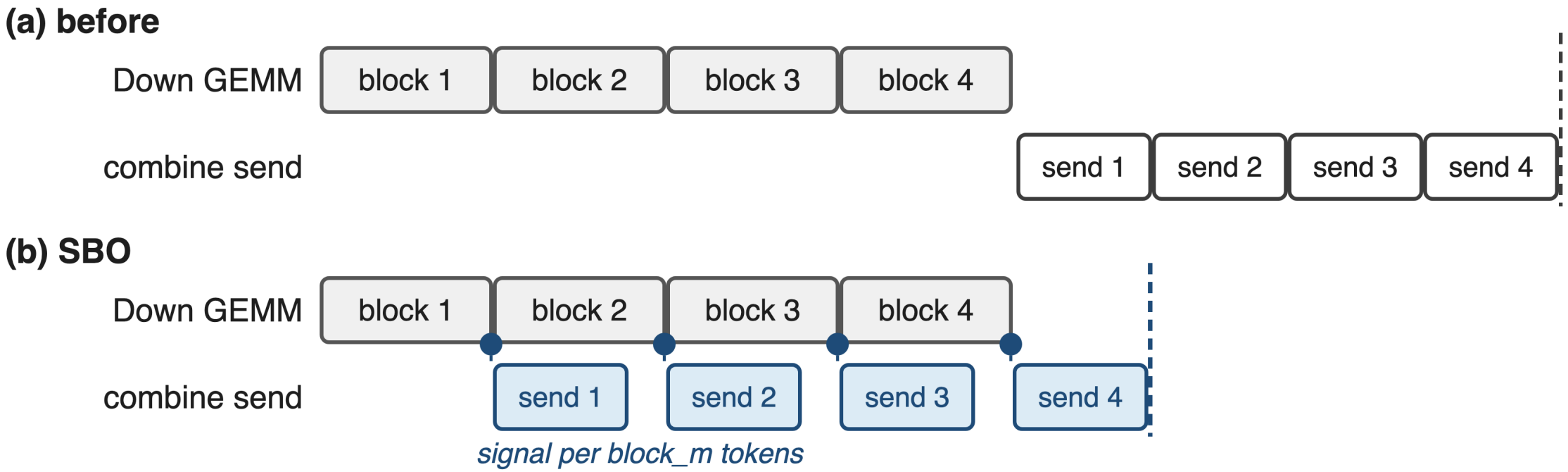
DeepEP V2: adaptive routing at every load, despite added latency

❸ SMs hand bytes to the NIC, then leave



Large RDMA buffer: send now, receive later through a hook (Normal-SMFree)

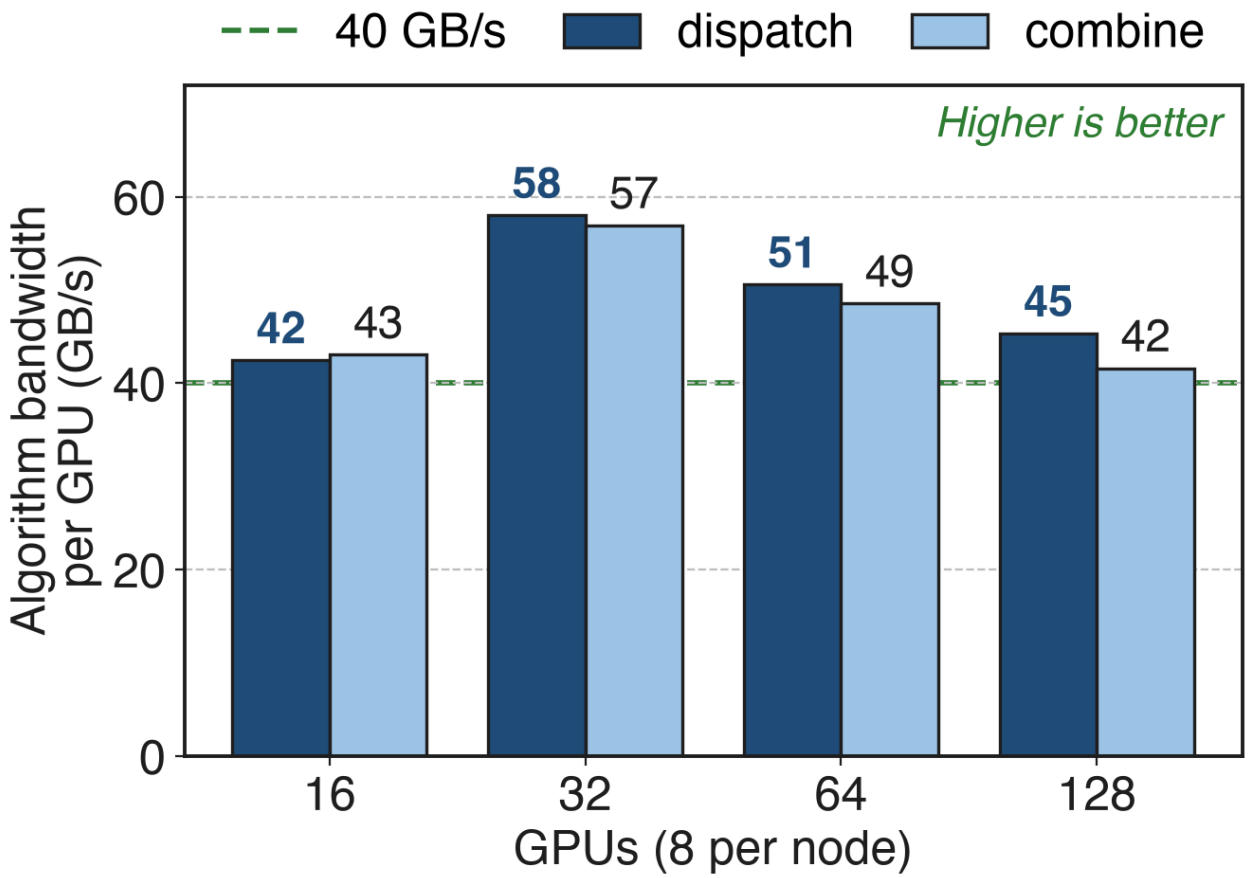
❸ SBO sends each GEMM block as it lands



LL-SBO: +8–10% decode throughput on 16 H20s at batch 20–56

Isolated planes keep EP above 40 GB/s per GPU

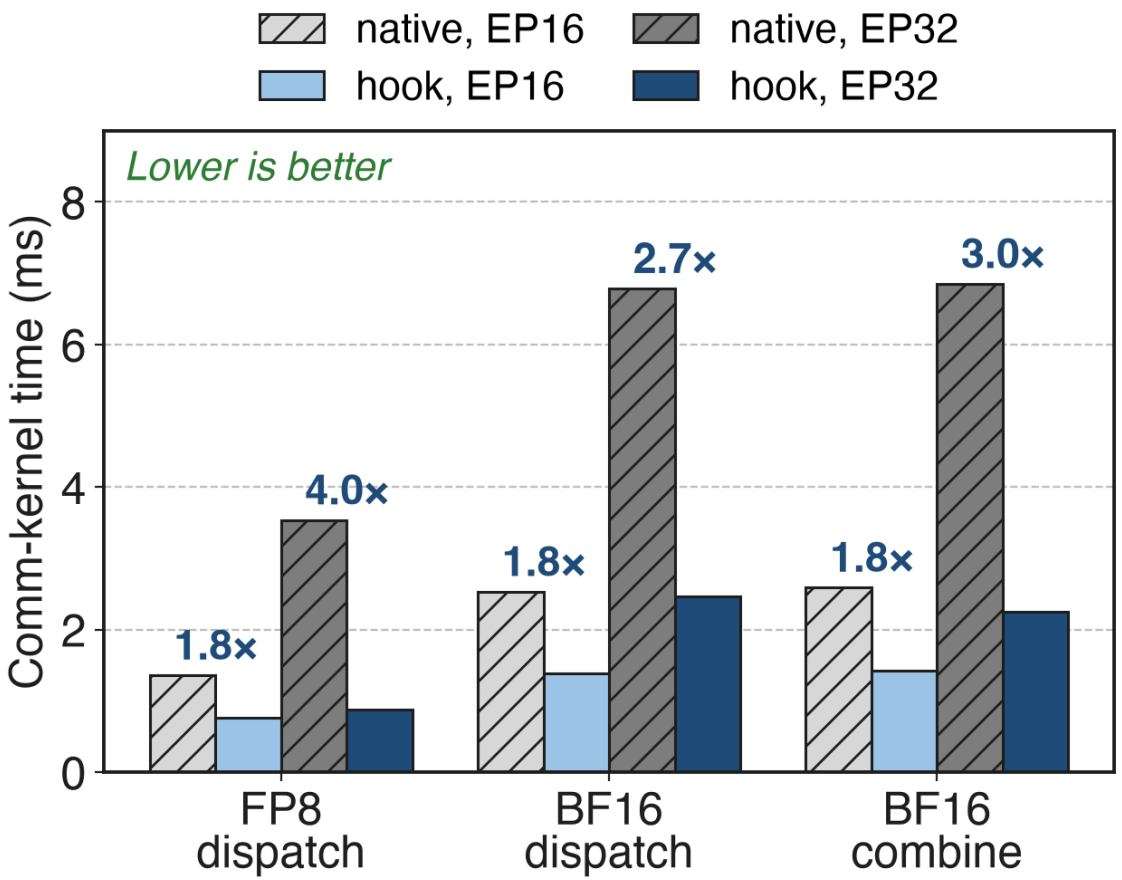
- DeepSeek's 8-plane IB fabric, 400 Gb/s NICs
- **4096** tokens per GPU, **16–128** GPUs
- Storage on its own RoCE plane



ISCA'25 Fig. 7: a tuned fabric end to end, not an A/B of VL or AR

Hook mode cuts comm-kernel time 1.8–4×

- **2–4** nodes × 8 H20, **4096** tokens, top-8
- Transfer overlapped by a GEMM
- Bigger gains at EP32



PR #347 (open): hook time counts only the send and receive kernels

The cost: fabric work, reordering, memory

Network

- SL→VL tables and AR are switch config
- AR reorders packets; Fire-Flyer chose static routing

SM-free

- ≈ **270** MB RDMA buffer per rank
- Mainline V2 keeps **4–6** SMs (was **24**)

Summary

- EP pays twice: shared fabric and **20 / 132** SMs
- **Insight**: all-to-all waits for its last packet; RDMA needs no SMs
- Isolated fabric: **> 40** GB/s per GPU at **16–128** GPUs
- SM-free hooks **1.8–4×**; SBO **+8–10%** decode

https://github.com/deepseek-ai/DeepEP